Conor McDonagh Rollo

Software Engineer | Cloud & Distributed Systems

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Profile

MSc Enterprise Software Systems graduate (June 2026) with a dissertation on carbon-aware batch job scheduling in Kubernetes, built around historical European grid carbon-intensity data from ENTSO-E. Production C#/C++ background as a Junior Game Programmer at Viridian Software, where I helped ship *Slay the Princess* to PlayStation, Xbox, and Nintendo Switch. Earlier experience in test automation, AWS, and ServiceNow. Looking for a software engineering role with a focus on cloud infrastructure, Kubernetes, or developer tools.

Education

MSc in Enterprise Software Systems | *South East Technological University, Waterford* Sept 2025 - June 2026 | Expected First-Class Honours (1.1)

- **Dissertation.** *Carbon-Aware Batch Job Scheduling in Kubernetes - A Simulation Study with Historical Grid Carbon-Intensity Data.* See Projects for detail.
- **Relevant modules.** Cloud Architecture, Cloud Application Services, Information Security, Enterprise Web Development, Agile Software Development, Mobile App Development.

BSc (Hons) Computer Games Development | *South East Technological University, Carlow* Sept 2020 - May 2024 | Second-Class Honours, Grade 1 (2.1)

- **Final year project.** OpenGL-based C/C++ game engine with procedural terrain generation. See Projects for detail.
- **Relevant modules.** Data Structures & Algorithms, Networking, 3D Graphics, Artificial Intelligence, Web Development & Databases, Agile Software Development, Assembly Programming.

Technical Skills

Cloud & DevOps	AWS (EC2, S3, CloudWatch, CodeBuild, Route53, Boto3), Kubernetes, Docker, Jenkins, Red Hat OpenShift, DigitalOcean (Droplets, doctl), Salesforce Heroku, ServiceNow, F5 BIG-IP
Languages	C++, C#, Python, Java, Kotlin, SQL, JavaScript, Lua
Frameworks & Tools	.NET Core, Flask, Pandas, Gradle, Maven, Git, OpenGL, Qt, Unity
Testing	Pytest, Google Test, NUnit, JUnit, KUnit, Selenium, test automation
Practices	Agile/Scrum, CI/CD, unit & integration testing, Jira, Rally

Experience

Junior Game Programmer | *Viridian Software, DCU Alpha, Dublin* July 2024 - Dec 2024 | C#, C++, Python

- Helped ship *Slay the Princess* to PlayStation, Xbox, and Nintendo Switch, working across the Microsoft GDK, Nintendo NDK, and PlayStation SDK toolchains.
- Ported core gameplay features to Switch and PlayStation in C++/SDL, including hardware gyroscope integration to drive a parallax effect on both platforms for use with Dualshock and JoyCons.
- Contributed system implementations and unit tests to a proprietary C# game engine and authored 100+ tests maintaining 70%+ code coverage.
- Debugged and resolved platform-specific issues across the multi-project console release.

Automation Test Engineer Intern | *UNUM, Carlow* March 2023 - Aug 2023 | Java, Selenium, AWS, ServiceNow

- Built Selenium test suites for a Java-based web application, verifying behaviour against AWS-hosted environments and logging defects for triage.
- Collaborated with other interns to automate the retirement of F5 load-balancer pools through ServiceNow, replacing a manual operational workflow and saving networking engineers 40+ hours per month.
- Participated in daily stand-ups, sprint retrospectives, and delivered weekly progress reports to the QA team.

Selected Projects

Carbon-Aware Kubernetes Scheduler Simulation MSc dissertation, 2025 - 2026

- Built a modular Python simulation framework for evaluating carbon-aware batch job scheduling on Kubernetes, comparing baseline, temporal deferral, spatial relocation, and combined strategies against historical grid carbon-intensity data from ENTSO-E across European regions in a simulated timeline.
- Components include a synthetic workload generator, a carbon time-series module, a policy engine, and a Kubernetes execution backend running BusyBox jobs to capture real timing telemetry.
- Decoupled carbon-aware decisions from the Kubernetes default scheduler so the framework remains reusable with swappable data sources and policies.

OpenGL Game Engine BSc final year project, 2023 - 2024

- Built an inheritance and polymorphism-based C/C++ engine on OpenGL, with base classes (`GameObject`, `Terrain`, `Input`) designed to support building games in environments without a GUI toolkit.
- Implemented a procedural terrain generator using layered noise octaves and a configurable detail value to produce levels from user parameters.